

Evaluation of a Carbonic Anhydrase Mimic for Industrial Carbon Capture

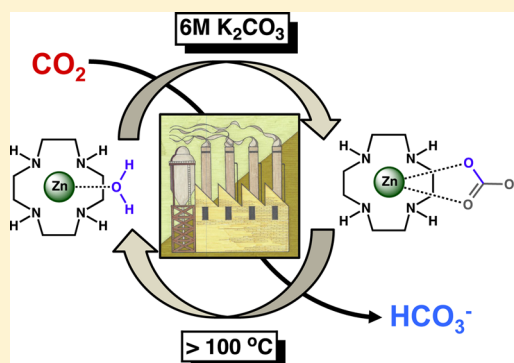
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S Supporting Information

ABSTRACT: Zinc(II) cyclen, a small molecule mimic of the enzyme carbonic anhydrase, was evaluated under rigorous conditions resembling those in an industrial carbon capture process: high pH (>12), nearly saturated salt concentrations (45% K₂CO₃) and elevated temperatures (100–130 °C). We found that the catalytic activity of zinc cyclen increased with increasing temperature and pH and was retained after exposure to a 45% w/w K₂CO₃ solution at 130 °C for 6 days. However, high bicarbonate concentrations markedly reduced the activity of the catalyst. Our results establish a benchmark level of stability and provide qualitative insights for the design of improved small-molecule carbon capture catalysts.



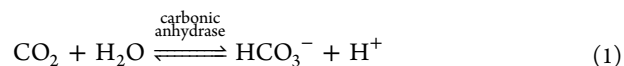
INTRODUCTION

Approximately 30 Gt of anthropogenic carbon dioxide are released to the atmosphere each year, leading to an accumulation of this greenhouse gas over time.^{1–3} Due to the risks associated with climate change, there is a growing international effort to develop commercially viable carbon capture technologies that can be implemented on a global scale.¹ Multiple avenues of approach have been pursued for low cost and large scale carbon capture, including membrane separation,⁴ solid sorbents,^{5–7} and mineralization,^{8,9} and these have been extensively reviewed elsewhere.^{10,11} Currently, the most established method of carbon capture from flue gas requires flowing an aqueous amine solution, such as monoethanolamine (MEA), in an absorber tower.¹² MEA rapidly absorbs CO₂, is inexpensive, and available in large quantities. However, the parasitic energy requirements of MEA-based carbon capture are estimated to be between 20 and 30% of the total energy produced by the plant, due to the high temperatures (>100 °C) required to remove CO₂ from the solution.^{1,13,14} Additionally, amine-based solvent systems release volatile organic compounds (VOCs),¹⁵ which are damaging to human health and the environment.¹⁶

Replacing MEA with aqueous carbonate solutions could significantly reduce the energy requirements for carbon capture, as carbonates require a lower temperature to release their stored CO₂. Additionally, carbonates are readily abundant, have low toxicity and do not emit VOCs. These attractive attributes make the utilization of carbonate solvents for carbon capture an active area of research.^{17–19} However, carbonate-based solvents

absorb carbon dioxide slowly; a carbonate process would need to be accelerated to be commercially viable. Fortunately, the rate of carbon capture in carbonate solvents can be increased through use of catalysts.^{20,21} Carbonic anhydrase (CA), for example, catalyzes the hydration of carbon dioxide as described by eq 1 with a turnover rate on the order of 10⁶ mol^{–1} s^{–1} at near neutral pH.¹⁴

CA catalyzed hydration of carbon dioxide



Despite unparalleled CO₂ hydration activity, CA enzymes are susceptible to thermal denaturation²² and are therefore expected to lose function at industrial absorption temperatures (40–60 °C) and likely degrade quickly at the even higher regeneration temperatures (100–120 °C). Enzymes are also sensitive to the highly alkaline environment found in industrial CO₂ sorption columns, where both denaturation and peptide hydrolysis can occur. Production of enzymes can be costly due to the difficulties of cell culture and enzyme purification and extraction from cellular material. In order to circumvent the

Received: April 17, 2013

Revised: June 19, 2013

Accepted: July 24, 2013

Published: July 24, 2013

potential limitations of CA and other enzymatic catalysts, here we explore the stability of a small molecule that was originally designed to mimic the active site of CA.²³ To our knowledge, the most active small molecule CA mimic identified thus far is the 1,4,7,10-tetraazacyclododecane chelate of zinc(II) perchlorate, or simply “zinc cyclen” (Figure 1), which will be the focus of this work.

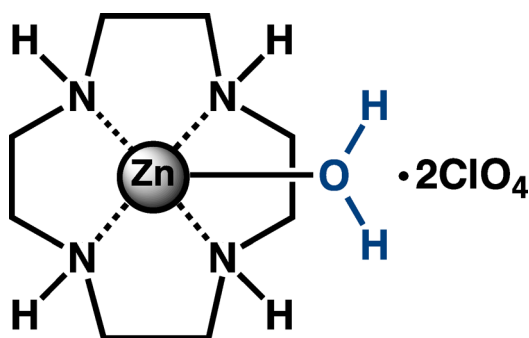


Figure 1. Zinc cyclen perchlorate catalyst used in this work which consists of a hydrated zinc atom coordinated to a cyclic amine ligand.

While zinc cyclen displays significantly slower CO₂ hydration kinetics²³ than CA, on a per mass basis zinc cyclen has only a 5-fold lower activity than CA due to its lower molecular weight (455 Da vs 30 000 Da). Additionally, the cyclen ligand is commercially available in large scale. Previous studies of small molecule CA mimics focused on comparing the mimic to the enzyme to uncover the CA mechanism; therefore, the small molecule mimics were usually evaluated only under the mild, physiologically relevant conditions optimal for CA.^{23,27,29} While the design of improved small molecule CA mimics for carbon capture is an important endeavor,^{24–30} it is equally important to ensure that this class of compounds will remain functional during industrial use. Here we establish the stability and activity of a benchmark small-molecule CA mimic under several critical end-use conditions.

■ EXPERIMENTAL METHODS

General. Unless otherwise noted, reagents and chemicals were purchased from VWR (Brisbane, CA, USA) or Sigma Aldrich (St Louis, MO, USA) and used without further purification. The cyclen ligand was purchased from Strem Chemicals (Newburyport, MA, USA). Zinc cyclen perchlorate was prepared as previously described.²³ ¹H and ¹³C NMR spectra were recorded at 30.0 ± 0.1 °C on a Bruker Avance III 600 MHz spectrometer (Bruker Biospin, Billerica, MA, USA) equipped with a 5 mm z-gradient broadband probe in D₂O containing 0.1% *t*-butanol as an internal reference standard (1.24 ppm).

Stopped-Flow Rate Constant Determination. Experimental catalytic rate constants for the CO₂ hydration reaction were determined using stopped-flow spectrophotometry using methods similar to those previously described.³⁰ Prior to the experiment, a solution of CO₂ saturated water was prepared by sparging deionized water with 100% CO₂ gas at 25 °C for at least 30 min. Using Henry's constant, this solution was calculated to contain 33.8 mM [CO₂]. Generally, unless otherwise noted, a solution containing 0.2 M NaClO₄, 0.1 M AMPSO (*N*-(1,1-dimethyl-2-hydroxyethyl)-3-amino-2-hydroxypropanesulfonic acid) buffer, and 5 × 10^{−5} M thymol blue indicator was evacuated under vacuum for 1 h followed by

sparging with nitrogen for 20 min to remove dissolved CO₂. A baseline uncatalyzed rate was determined by rapidly mixing the dissolved CO₂ solution and an equal volume of the buffer solution in an Applied Photophysics stopped-flow spectrophotometer while recording the time-dependent absorbance at λ = 596 nm. Solutions containing 1–2 mM zinc cyclen perchlorate were prepared by stirring under nitrogen. Initial rates for each catalyst concentration were calculated by fitting the time-dependent absorbance data with a single exponential decay function. The subset of data used in the fit corresponded to 10% the total reaction time (the time required for the absorbance to reach its equilibrium value). Each catalyst and buffer solution was measured eight times, and the standard deviation determined. Initial rates were calculated using the equation $v_{\text{init}} = Q(A_0 - A_e)[d(\ln(A - A_e))/dt]_{t \rightarrow 0}$ where Q is the buffer factor and A_0 and A_e were the initial and final absorbance values, respectively. The value of Q was determined by mixing the buffer solution 1:1 with three different HCl/water solutions and measuring the resulting solution absorbance in the stopped flow spectrophotometer. The HCl concentrations measured were 0.0076, 0.0153, and 0.031 M and were chosen to represent a range of [H⁺] similar to the [H⁺] generated in the CO₂ hydration reaction. The rate constant k_{cat} was determined as the slope of $v_{\text{init}}/[CO_2]$ vs [zinc cyclen (ClO₄)₂], and the error was determined as the standard deviation among replicates. For the thermal stability experiments shown in Figure 4, 0.1 M catalyst was incubated in 1 M K₂CO₃ for the time period indicated, followed by dilution 1:100 in 0.1 M AMPSO buffer prior to measurement. For the bicarbonate titration experiments shown in Figure 6, 0.25 M TAPS buffer (*N*-[tris(hydroxymethyl)methyl]-3-aminopropanesulfonic acid) with *m*-cresol purple pH indicator was acidified to pH 7.5 using HClO₄ prior to addition of KHCO₃. Subsequently, the pH of the final solutions were adjusted to 8.25 using KOH. The buffer factor was measured for each KHCO₃ concentration. The reaction progress was monitored at λ = 578 nm.

Solvation of Zinc Carbonate. To 1.48 g (11.8 mmol) of solid ZnCO₃ in 10 mL of deionized water was added 2.06 g (11.9 mmol) of the cyclen ligand. The solution was agitated with mild heating until a homogeneous solution was obtained (approximately 5 min). The total volume of the resulting solution was approximately 12 mL. Removal of the water under reduced pressure yielded the product in quantitative yield. The ¹H NMR for this material was identical to that of the zinc cyclen complex prepared by published literature methods.³¹

Precipitation Experiments. To a homogeneous 2 mL solution of 1 M K₂CO₃ was added 1 mL of a 0.2 M homogeneous solution of zinc cyclen perchlorate. A white, insoluble precipitate was immediately formed and isolated by filtration, washed with a small amount of cold water, and dried under reduced pressure. Identification of the solid was achieved by EDS using a JEOL 7300 SEM equipped with an Oxford Inca microanalysis system. Prior to measurement, the precipitates (unknown, along with zinc cyclen, ZnCO₃, and KClO₄ controls) were dried in air and deposited on a carbon tape.

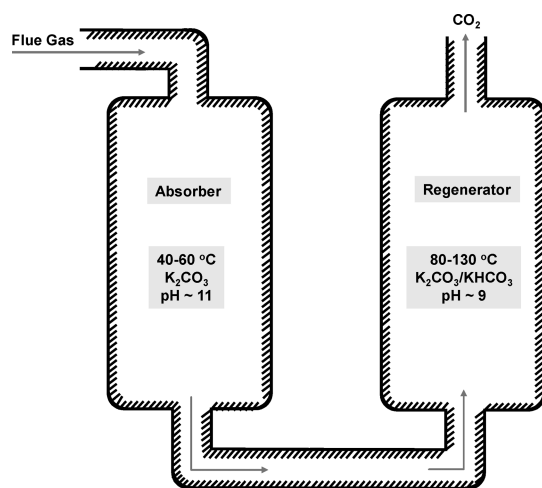
High Temperature and Carbonate Survivability. To a solution of 45 wt % K₂CO₃ in D₂O (1 mL) was added zinc cyclen perchlorate (45 mg) and 0.1% wt/vol *t*-butanol as an internal reference standard. The solution was then agitated for several minutes, filtered over Celite, and placed in an NMR tube. The sample was then incubated in a 130 °C oil bath for 6 days and monitored daily by ¹H NMR.

Pressure Drop Experiments. The pressure-drop apparatus was constructed based upon a published design.³² Briefly, the custom 50 mL volume, water-jacketed glass reactor (4 cm diameter, 4 cm tall) was constructed to fit an O-ring sealed stainless steel vacuum flange as the lid. The flange was welded with a port for a 4-way cross that is connected to a pressure transducer (Omegadyne no. MMA015USBP6COT8A9), a pressure relief device, and a gas inlet. The reactor was fitted with three baffles to aid in mixing. In a typical measurement, the reactor temperature was set to 25 °C, filled with a 10 mL test solution, and evacuated to 0.4–0.45 psia (the vapor pressure of the liquid at 25 °C) using a diaphragm pump (Cole Parmer no. NO35.3 TTP). The solution was stirred at 300 rpm, and 100% CO₂ was pumped into the reactor through a needle valve to reach a pressure of 2 psia. The chamber was then sealed, and the change in pressure versus time (which was linear) was recorded for 30 s. The rate of CO₂ sorption was determined as the slope of the pressure vs time trace. The error shown in Figure 5 is the standard deviation among four replicates.

RESULTS AND DISCUSSION

The industrial process and conditions which dictated our lab-scale experiments are shown in Scheme 1 and are as follows: In

Scheme 1. Simplified Industrial Carbon Capture Process: Initial Conditions in the Absorber and the Regenerator

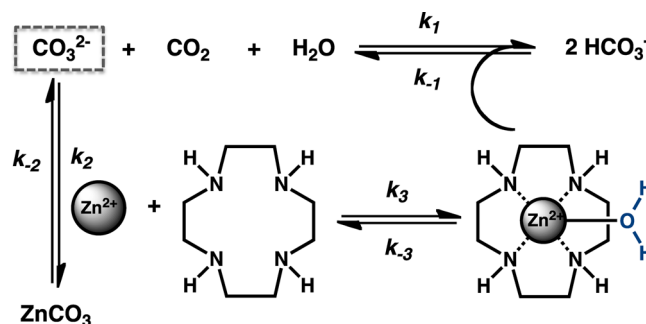


a coal fired power plant, the flue gas stream is passed through a gas–liquid contactor containing solvent heated to 40–60 °C. The solvent (buffer) concentration is set just below saturation level to maximize carrying capacity. For potassium carbonate-based solvent systems, the solubility limit is approximately 45% wt/wt, or about 6 mol/L, and we assumed a pH swing between approximately 11 and 9. After loading, the solvent is taken to a desorption column which is heated to higher temperatures (80–130 °C) to release stored CO₂. To test the catalyst under the most stringent thermal conditions, we assumed a regeneration temperature of 130 °C, which represents the upper end of amine solvent regeneration temperatures.

Previous studies of the zinc cyclen catalyzed CO₂ hydration reaction have focused solely on the kinetics and mechanism and have therefore been performed under conditions with intentionally low carbonate (CO₃²⁻) and bicarbonate (HCO₃⁻) concentrations in order to simplify the system to the reaction shown in eq 1. However, in focusing on the industrial

application of zinc cyclen in potassium carbonate solvent we must consider the chemical reactions and product inhibition that may occur due to significantly higher CO₃²⁻ and HCO₃⁻ concentrations. For example, as shown in Scheme 2, CO₃²⁻

Scheme 2. Potential Chemical Reactions between Zinc Cyclen and Carbonate/Bicarbonate Anions



(boxed) can react with Zn²⁺ to form insoluble ZnCO₃ in water with a rate of k_2 ; loss of zinc from the catalyst active site is a potential route to catalyst inactivation in high CO₃²⁻ concentrations. Additionally, the reaction between CO₂ and water in carbonate solvent yields two HCO₃⁻ molecules at a rate of k_1 , leading to two times higher HCO₃⁻ concentrations in carbonate solvents during CO₂ loading than in other (e.g. amine) solvents. While the accumulation of high concentrations of bicarbonate likely increases the rate of the backward reaction (k_{-1}), the extent of this “bicarbonate inhibition” in a practical catalyzed system has not been evaluated prior to our study.

We systematically evaluated the stability of zinc cyclen under several expected industrial conditions by monitoring the structure and activity under the thermal and chemical conditions shown in Schemes 1 and 2. We tested the thermal and chemical conditions separately to isolate potential routes of degradation.

Catalyst Thermal Stability and Activity: Absorber Column. As illustrated in Scheme 1, the two phases of solvent-based carbon capture are (1) CO₂ absorption (left) and (2) solvent regeneration (right). The absorber column reaches 40–60 °C during the CO₂ absorption phase. Using a stopped flow spectrophotometer equipped with syringes and seals that could withstand temperatures up to 75 °C, we directly measured the zinc cyclen catalyzed initial rates in situ from 25 to 75 °C. As described in the Methods section, the stopped-flow derived rate constants were determined using a colorimetric readout using buffer/pH indicator pairs with matching pK_a ,³³ and we performed this experiment in AMPSO buffer. As shown in Figure 2, as we increased the temperature, the catalyst showed a significant enhancement in activity relative to the background rates, with an increase in turnover rate from 3000 mol⁻¹ s⁻¹ to approximately 70 000 mol⁻¹ s⁻¹. In contrast, in a separate experiment (not shown) we found that CA lost all activity at 55 °C, consistent with literature reports.²² Our results indicate that the high temperatures present in the absorber column will not degrade zinc cyclen and, furthermore, will enhance catalytic activity.

To determine whether zinc cyclen withstands the higher temperatures in the regenerator, a sample was refluxed in D₂O (100 °C) for five days, and the structure was monitored using ¹H NMR spectroscopy. We did not detect any structural

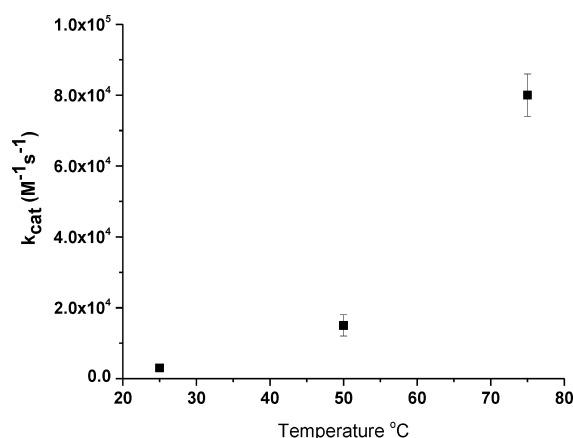


Figure 2. Zinc cyclen activity (observed rate constant in units of per molar per second) at 25, 50, and 75 °C. Catalyst activity increases significantly with increasing temperature.

change in the catalyst during the thermal treatment (see the Supporting Information), establishing that zinc cyclen does not chemically decompose at elevated temperatures and would likely remain intact in the regenerator in an industrial carbon capture facility (in the absence of solvent effects).

Catalyst Stability in Carbonate Solutions. We concurrently investigated the chemical compatibility of zinc cyclen with carbonate buffers. As shown in Scheme 2, the zinc cyclen complex is in equilibrium with Zn^{2+} and the cyclen ligand. If the CO_3^{2-} present in the system reacts with the free Zn^{2+} , Zn^{2+} will be quantitatively removed from the system as solid $ZnCO_3$, as its solubility is low enough to be negligible (<1 mg/L).³⁴ Therefore, we performed solubility studies to determine whether Zn(II) remains bound to the cyclen ligand in the presence of high concentrations of carbonate.

We mixed a 0.1 M zinc cyclen solution with a 1 M potassium carbonate solution at room temperature. A white solid immediately precipitated which was identified by EDX as potassium perchlorate, formed by the combination of the potassium ion with the perchlorate counterion used to prepare the catalyst (see the Supporting Information). When we performed the same experiment using catalyst prepared as the sulfate or chloride salt rather than the perchlorate, or when sodium carbonate was used in place of potassium carbonate, precipitation was not observed. The observation that no zinc carbonate precipitate was formed after incubation of zinc cyclen in concentrated potassium carbonate solution suggests that the cyclen ligand can prevent immediate precipitation of zinc carbonate.

To more rigorously evaluate the relative affinity of Zn(II) for cyclen vs CO_3^{2-} in carbonate solutions, we combined stoichiometric amounts of solid zinc carbonate and cyclen in water. After several minutes of agitation and mild heating (the reaction is mildly exothermic), we obtained a clear solution of zinc cyclen. After removal of the solvent, ¹H NMR and catalytic activity of the solid obtained was identical to that of the zinc cyclen catalyst prepared as the perchlorate. Collectively, our results indicate that the cyclen ligand has a higher affinity than does carbonate for Zn(II); therefore the zinc cyclen catalyst remains intact in concentrated potassium carbonate solvent.

Regenerator Conditions: Activity Effects. We found that zinc cyclen remained active under the thermal conditions in the absorber tower, remained structurally intact in the higher temperatures in the regenerator, and did not precipitate zinc

carbonate in concentrated potassium carbonate solutions. To investigate whether zinc cyclen remained structurally intact after exposure to regenerator column conditions, which consist of heated, concentrated potassium carbonate, we incubated zinc cyclen in 6 M (approximately 45 wt %) potassium carbonate solutions at 130 °C for 6 days. We detected no structural changes by ¹H NMR analysis (see the Supporting Information). In contrast, the hot potassium carbonate conditions began to dissolve the glass in the NMR tubes and ended the experiment. While the NMR experiment indicated that zinc cyclen remained structurally intact, we wanted to verify that the structural data corresponded to retained activity. Due to the high temperatures involved and the lack of an appropriate pH indicator we could not measure zinc cyclen activity directly under regenerator conditions in situ, therefore we incubated the zinc cyclen at 100 °C in 1 M potassium carbonate and removed aliquots for testing. After cooling, dilution, and pH adjustment, we performed stopped flow analysis of these aliquots. The relationship between catalytic activity and incubation time after incubation of zinc cyclen in the thermal and chemical conditions found in the regenerator column is shown in Figure 3. The figure shows that catalyst retained 100% of its catalytic activity during this incubation.

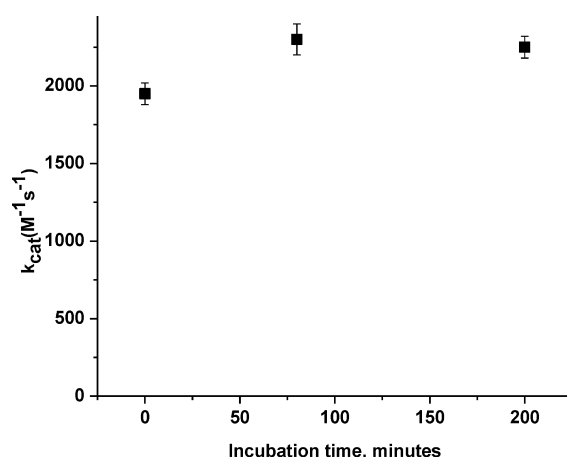


Figure 3. Kinetic activity (observed rate constant, in units of per molar per second) as a result of incubation time in regenerator conditions (1 M potassium carbonate, pH 11.0, 100 °C).

Catalytic Activity in Concentrated Carbonate Solutions. Our results demonstrate that zinc cyclen is structurally intact after exposure to conditions in a potassium carbonate absorber and regenerator column. While stopped-flow spectrophotometry enables the direct measurement of the overall reaction rate, stopped flow is most accurate when using moderate buffer concentrations and pH ranges, and when the buffer and indicator have similar pK_a values.³³ Because we could not identify a water-soluble pH indicator with a pK_a matched to the that of the bicarbonate anion (10.2), we used a pressure-based rather than a colorimetric method to determine zinc cyclen activity directly in concentrated alkaline solutions. This “pressure drop” method³² measures the flux of CO_2 as a decrease in CO_2 pressure above the solution in a sealed reactor. As shown in Figure 4, 50 mM zinc cyclen enhances the flux of CO_2 in 0.1, 1, and 3 M potassium carbonate solutions. However, as the carbonate concentration increases from 0.1 (black line) to 3 M (blue line), the effect of zinc cyclen on the rate of CO_2 absorption is greatly diminished. The diminished

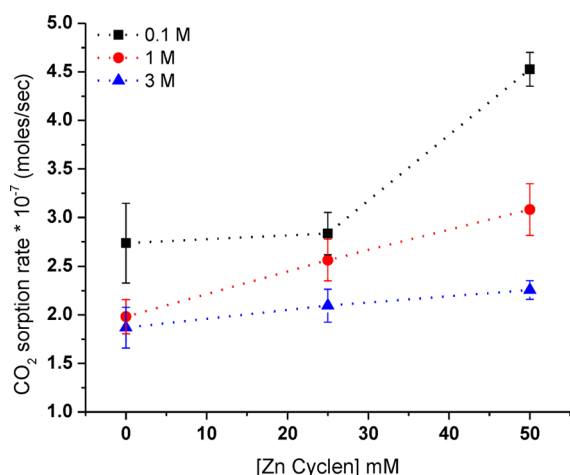


Figure 4. CO₂ sorption rate in carbonate buffer between 0.1 and 3 M potassium carbonate at pH 9.9 and 25 °C. Reaction rates decrease with increasing carbonate concentration and increase with increasing catalyst concentration.

apparent catalytic activity with increasing carbonate buffer concentrations indicates that zinc cyclen is inhibited in potassium carbonate solutions. Strong bicarbonate and carbonate inhibition have been observed for some isoforms of CA^{35,36} and may represent a considerable challenge in catalyzed CO₂ capture. However, the fact that some CA isoforms are inhibited to a lesser extent than others³⁵ suggests that CA mimics might be designed to minimize bicarbonate and carbonate inhibition.

Our results shown in Figure 4 indicate that zinc cyclen exhibits diminished activity in highly concentrated potassium carbonate solutions (which contain both carbonate and bicarbonate) at pH 9.9. To determine the extent to which specifically the bicarbonate species inhibits zinc cyclen, we measured the catalyzed reaction rate at different concentrations of bicarbonate while maintaining constant pH (Figure 5). As we increased the bicarbonate concentration, the reaction rate decreased dramatically, falling to the baseline uncatalyzed rate in 0.7 M bicarbonate, strongly suggesting that catalyst inhibition by bicarbonate contributes to the anomalously low catalyst activity seen in carbonate buffer in the experiments

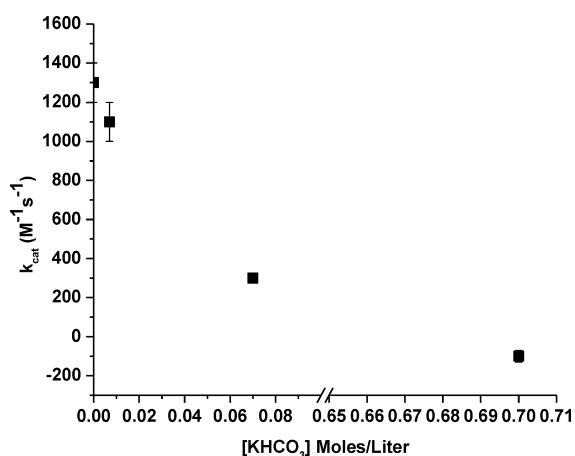


Figure 5. Observed rate constants, in units of per molar per second, measured for 1 mM zinc cyclen as a function of bicarbonate concentration at pH 8.3 and 25 °C.

shown in Figure 4. The result is consistent with our previous combined computational and experimental study which suggested that release of bicarbonate from the catalyst active site is the rate-limiting step in the zinc cyclen catalyzed hydration of CO₂.³⁰ It follows that high concentrations of bicarbonate in the system increases the rate of the backward reaction, (k^{-1} in Scheme 2) significantly impeding the overall reaction rate.

Despite the evidence of strong bicarbonate inhibition of zinc cyclen shown in Figure 5, the bicarbonate species is present in lower concentrations than the carbonate species in much of the pH range found in the absorber column (the pK_a of $HCO_3^- \rightarrow H^+ + CO_3^{2-}$ is 10.2). To understand the performance of zinc cyclen in carbonate buffers over this pH range (pH 11 to 9), we measured the flux of CO₂ in catalyzed potassium carbonate/bicarbonate solutions at varying pH. As shown in Figure 6, as

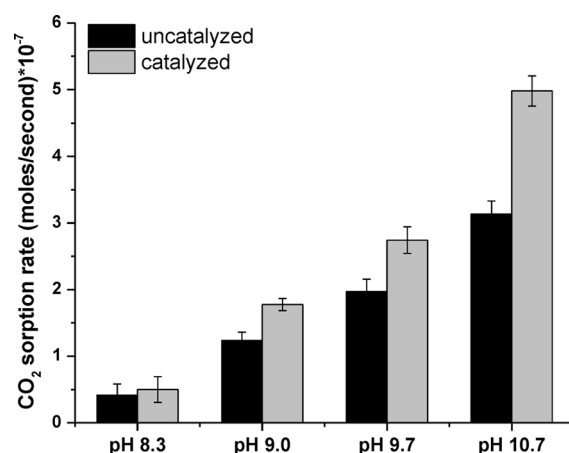


Figure 6. Pressure drop analyses of CO₂ uptake in 10 wt % (1 M) KHCO₃/K₂CO₃ solutions with and without 50 mM catalyst at different pH values. Catalyst activity increases with increasing pH.

the pH of the carbonate solution was increased (and the ratio of carbonate to bicarbonate increased) the ratio of catalyzed (gray) to uncatalyzed (black) CO₂ sorption rates also increased. The results indicate that carbonate solvent inhibition can be mitigated by operating at higher pH and that the bicarbonate anion may be a stronger inhibitor of zinc cyclen than the carbonate anion.

Trade-offs between activity, stability, and degree of bicarbonate inhibition will likely arise with new catalyst development, and the optimal choice of catalyst will depend on the overall process (temperatures and solvents). For example, we have developed a new class of acyclic CA mimics (unpublished) that exhibit slightly higher catalytic activity and are less inhibited by bicarbonate than zinc cyclen. However, the new catalysts do not survive the regenerator conditions described here. We did not address all possible process variables in the present study, such as the wide differences in the composition of flue gas (including varying compositions of SO_x and NO_x) from different sources.³⁷ The presence of impurities in the flue gas may also need to be considered in catalyst design and selection.

Our results suggest that zinc cyclen can serve as a robust catalyst for industrial carbon capture, as its activity increases with both increasing temperature and pH. Zinc cyclen is likely one of the most stable CA mimics possible because the cyclen ligand employed in the zinc cyclen catalyst is a highly stable

macrocyclic metal chelator as described by the “chelate effect”.³⁸ However, zinc cyclen activity is diminished at low pH (<9) or in the presence of high (>0.5 M) bicarbonate concentrations. Since bicarbonate has been shown to inhibit both CA and zinc cyclen, inhibition also likely affects other small molecule CA mimics. Additionally, bicarbonate inhibition of carbon capture catalysts will also likely occur in other (e.g., hindered and tertiary amine) catalyzed solvent systems where bicarbonate concentrations will increase to >1 M during CO₂ loading. Therefore, routes for circumventing inhibition of carbon capture catalysts are broadly required. While some variants of CA minimize bicarbonate inhibition by protecting the active site with a hydrophobic pocket, other approaches to catalyst design may include using an anionic ligand to repel bicarbonate and other potential anionic inhibitors. System design for zinc cyclen may include employing carbonate species with less soluble bicarbonates or limiting the industrial swing cycle to include only high pH environments.

■ ASSOCIATED CONTENT

■ Supporting Information

Additional data including NMR spectra of heated zinc cyclen, EDS analysis of precipitates, and stopped flow analysis of the catalyst. This information is available free of charge via the Internet at <http://pubs.acs.org/>.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344. Funding provided by Advanced Research Projects Agency–Energy, US Department of Energy Innovative Materials and Processes for Advanced Carbon Capture Technology (IMPACCT) Program Award 09/CJ000/05/01

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